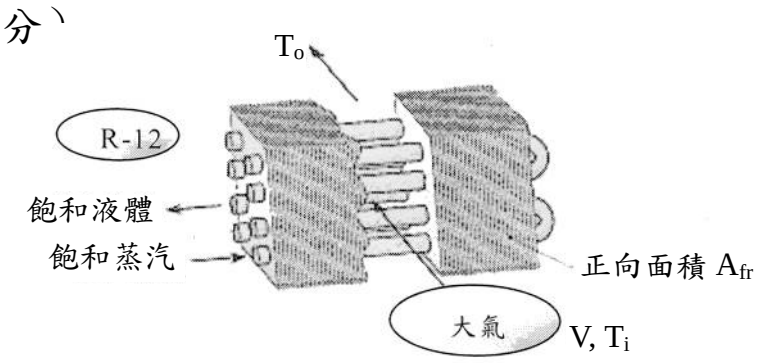
九十二年專門職業及技術人員律師、會計師、建築師、技師<br>社會工作師、土地登記專業代理人檢覈筆試試題　　代號：31210　　全三頁<br>第一頁

類　　科：冷凍空調工程技師
科　　目：冷凍空調設計
考試時間：二小時　　座號：＿＿＿＿＿＿

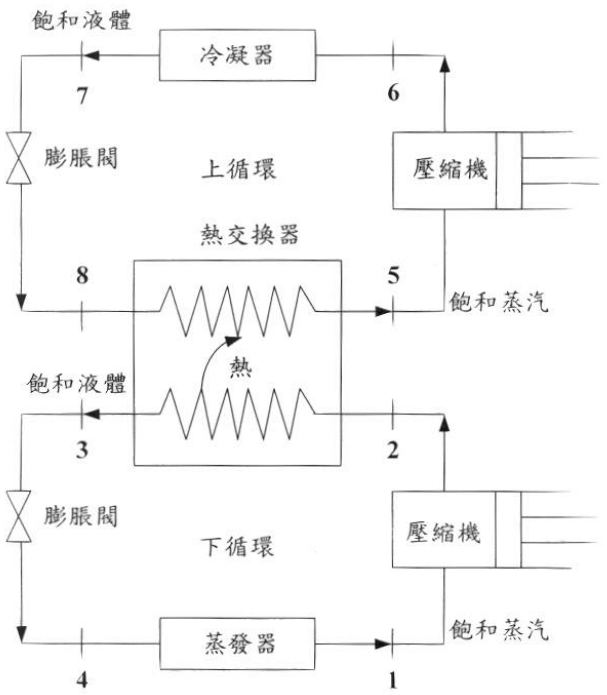
※注意：㈠不必抄題，作答時請將試題題號及答案依照順序寫在試卷上，於本試題上作答者，不予計分。
㈡可以使用電子計算器，但需詳列解答過程。

一、在冷凍空調系統中常使用氣冷式板鰭管熱交換器來冷凝 R-12 飽和蒸汽，如下圖所示。
R-12 的飽和溫度為 45℃，冷凝量為 0.015 kg/s，其潛熱 $h_{fg}$ 為 135 KJ/kg。冷凝器正向面積 $A_{fr}$ 固定為 0.25$m^2$，空氣入口溫度 $T_i$ 為 30℃，入口速度 V 為 2 m/s，熱交換器空氣側的總括熱傳係數 U 為 50 W/$m^2$，求：
㈠要達到上述的冷凝量需要多少熱傳表面積($m^2$)。（15 分）
㈡空氣出口溫度（℃）。（10 分）




註：空氣之定壓比熱 $C_p$=1.004 KJ/kg K，密度為 1.17 kg/$m^3$

二、考慮一個兩段式串聯(two-stage cascade)冷凍系統，如右下圖所示。其最高壓力為 0.8 MPa，最低壓力為 0.14 MPa，以冷媒 R-134a 為工作流體，進行理想的蒸汽壓縮冷凍式循環。下循環(lower cycle)與上循環(upper cycle)之冷媒經由一個絕熱熱交換器進行熱交換，其冷媒壓力均為 0.4 MPa，假設冷媒經過上循環的質量流率是 0.24 kg/s，求：
㈠繪出此循環之 T-S(temperature-entropy)圖。（5 分）
㈡冷媒經過下循環的質量流率(kg/s)。（5 分）
㈢此冷凍機的性能係數(coefficient of performance)。（15 分）




三、有一個空調系統包括一個加熱器及蒸發冷卻器，空氣進入加熱器前，溫度 10℃，相對濕度 70%，體積流率 30 $m^3$/min。離開蒸發冷卻器時，其溫度為 20℃，相對濕度 60%，求：
㈠空氣離開加熱器時之溫度（℃）及相對溼度（%）。（8 分）
㈡加熱器所需之熱功率 (KJ/min)。（9 分）
㈢蒸發冷卻器所需之噴水量(kg/min)。（8 分）

四、有一個邊長 3m 長之正立方體冷凍庫，其外壁面為厚度 6.35 mm 之鋼板（熱傳導係數 k=64 W/m K），內壁面為厚度 6.35 mm 之鋁板(k=173 W/m K)。中間充填絕熱材料其厚度為 100 mm (k=0.039 W/m K)。假設絕熱材料與鋼板及鋁板之間的粘著劑其接觸熱阻均為 2.5×$10^{-4}$$m^2$ K/W，若冷凍庫之內外壁溫度欲保持為-6℃及 22℃時，則冷凍庫之冷凍能力(cooling load)應為多少(W)？（25 分）

（請接第二頁）

類　　科：冷凍空調工程技師
科　　目：冷凍空調設計

附表一：冷媒 R-134a 飽和特性表

Saturated refrigerant-134a—Pressure table

| Press., $P$ MPa | Temp., $T_{sat}$ °C | Specific volume, m³/kg: Sat. liquid, $v_f$ | Specific volume, m³/kg: Sat. vapor, $v_g$ | Internal energy, kJ/kg: Sat. liquid, $u_f$ | Internal energy, kJ/kg: Sat. vapor, $u_g$ | Enthalpy, kJ/kg: Sat. liquid, $h_f$ | Enthalpy, kJ/kg: Evap., $h_{fg}$ | Enthalpy, kJ/kg: Sat. vapor, $h_g$ | Entropy, kJ/kg · K: Sat. liquid, $s_f$ | Entropy, kJ/kg · K: Sat. vapor, $s_g$ |
|---|---|---|---|---|---|---|---|---|---|---|
| 0.06 | −37.07 | 0.0007097 | 0.3100 | 3.41 | 206.12 | 3.46 | 221.27 | 224.72 | 0.0147 | 0.9520 |
| 0.08 | −31.21 | 0.0007184 | 0.2366 | 10.41 | 209.46 | 10.47 | 217.92 | 228.39 | 0.0440 | 0.9447 |
| 0.10 | −26.43 | 0.0007258 | 0.1917 | 16.22 | 212.18 | 16.29 | 215.06 | 231.35 | 0.0678 | 0.9395 |
| 0.12 | −22.36 | 0.0007323 | 0.1614 | 21.23 | 214.50 | 21.32 | 212.54 | 233.86 | 0.0879 | 0.9354 |
| 0.14 | −18.80 | 0.0007381 | 0.1395 | 25.66 | 216.52 | 25.77 | 210.27 | 236.04 | 0.1055 | 0.9322 |
| 0.16 | −15.62 | 0.0007435 | 0.1229 | 29.66 | 218.32 | 29.78 | 208.18 | 237.97 | 0.1211 | 0.9295 |
| 0.18 | −12.73 | 0.0007485 | 0.1098 | 33.31 | 219.94 | 33.45 | 206.26 | 239.71 | 0.1352 | 0.9273 |
| 0.20 | −10.09 | 0.0007532 | 0.0993 | 36.69 | 221.43 | 36.84 | 204.46 | 241.30 | 0.1481 | 0.9253 |
| 0.24 | −5.37 | 0.0007618 | 0.0834 | 42.77 | 224.07 | 42.95 | 201.14 | 244.09 | 0.1710 | 0.9222 |
| 0.28 | −1.23 | 0.0007697 | 0.0719 | 48.18 | 226.38 | 48.39 | 198.13 | 246.52 | 0.1911 | 0.9197 |
| 0.32 | 2.48 | 0.0007770 | 0.0632 | 53.06 | 228.43 | 53.31 | 195.35 | 248.66 | 0.2089 | 0.9177 |
| 0.36 | 5.84 | 0.0007839 | 0.0564 | 57.54 | 230.28 | 57.82 | 192.76 | 250.58 | 0.2251 | 0.9160 |
| 0.4 | 8.93 | 0.0007904 | 0.0509 | 61.69 | 231.97 | 62.00 | 190.32 | 252.32 | 0.2399 | 0.9145 |
| 0.5 | 15.74 | 0.0008056 | 0.0409 | 70.93 | 235.64 | 71.33 | 184.74 | 256.07 | 0.2723 | 0.9117 |
| 0.6 | 21.58 | 0.0008196 | 0.0341 | 78.99 | 238.74 | 79.48 | 179.71 | 259.19 | 0.2999 | 0.9097 |
| 0.7 | 26.72 | 0.0008328 | 0.0292 | 86.19 | 241.42 | 86.78 | 175.07 | 261.85 | 0.3242 | 0.9080 |
| 0.8 | 31.33 | 0.0008454 | 0.0255 | 92.75 | 243.78 | 93.42 | 170.73 | 264.15 | 0.3459 | 0.9066 |
| 0.9 | 35.53 | 0.0008576 | 0.0226 | 98.79 | 245.88 | 99.56 | 166.62 | 266.18 | 0.3656 | 0.9054 |
| 1.0 | 39.39 | 0.0008695 | 0.0202 | 104.42 | 247.77 | 105.29 | 162.68 | 267.97 | 0.3838 | 0.9043 |
| 1.2 | 46.32 | 0.0008928 | 0.0166 | 114.69 | 251.03 | 115.76 | 155.23 | 270.99 | 0.4164 | 0.9023 |

附表二：冷媒 R-134a 過熱蒸汽性質表

Superheated refrigerant-134a

| $T$ °C | $v$ m³/kg | $u$ kJ/kg | $h$ kJ/kg | $s$ kJ/kg · K | $v$ m³/kg | $u$ kJ/kg | $h$ kJ/kg | $s$ kJ/kg · K | $v$ m³/kg | $u$ kJ/kg | $h$ kJ/kg | $s$ kJ/kg · K |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | $P$ = 0.06 MPa ($T_{sat}$ = −37.07°C) | | | | $P$ = 0.10 MPa ($T_{sat}$ = −26.43°C) | | | | $P$ = 0.14 MPa ($T_{sat}$ = −18.80°C) | | | |
| Sat. | 0.31003 | 206.12 | 224.72 | 0.9520 | 0.19170 | 212.18 | 231.35 | 0.9395 | 0.13945 | 216.52 | 236.04 | 0.9322 |
| −20 | 0.33536 | 217.86 | 237.98 | 1.0062 | 0.19770 | 216.77 | 236.54 | 0.9602 | | | | |
| −10 | 0.34992 | 224.97 | 245.96 | 1.0371 | 0.20686 | 224.01 | 244.70 | 0.9918 | 0.14549 | 223.03 | 243.40 | 0.9606 |
| 0 | 0.36433 | 232.24 | 254.10 | 1.0675 | 0.21587 | 231.41 | 252.99 | 1.0227 | 0.15219 | 230.55 | 251.86 | 0.9922 |
| 10 | 0.37861 | 239.69 | 262.41 | 1.0973 | 0.22473 | 238.96 | 261.43 | 1.0531 | 0.15875 | 238.21 | 260.43 | 1.0230 |
| 20 | 0.39279 | 247.32 | 270.89 | 1.1267 | 0.23349 | 246.67 | 270.02 | 1.0829 | 0.16520 | 246.01 | 269.13 | 1.0532 |
| 30 | 0.40688 | 255.12 | 279.53 | 1.1557 | 0.24216 | 254.54 | 278.76 | 1.1122 | 0.17155 | 253.96 | 277.97 | 1.0828 |
| 40 | 0.42091 | 263.10 | 288.35 | 1.1844 | 0.25076 | 262.58 | 287.66 | 1.1411 | 0.17783 | 262.06 | 286.96 | 1.1120 |
| 50 | 0.43487 | 271.25 | 297.34 | 1.2126 | 0.25930 | 270.79 | 296.72 | 1.1696 | 0.18404 | 270.32 | 296.09 | 1.1407 |
| 60 | 0.44879 | 279.58 | 306.51 | 1.2405 | 0.26779 | 279.16 | 305.94 | 1.1977 | 0.19020 | 278.74 | 305.37 | 1.1690 |
| 70 | 0.46266 | 288.08 | 315.84 | 1.2681 | 0.27623 | 287.70 | 315.32 | 1.2254 | 0.19633 | 287.32 | 314.80 | 1.1969 |
| 80 | 0.47650 | 296.75 | 325.34 | 1.2954 | 0.28464 | 296.40 | 324.87 | 1.2528 | 0.20241 | 296.06 | 324.39 | 1.2244 |
| 90 | 0.49031 | 305.58 | 335.00 | 1.3224 | 0.29302 | 305.27 | 334.57 | 1.2799 | 0.20846 | 304.95 | 334.14 | 1.2516 |
| 100 | | | | | | | | | 0.21449 | 314.01 | 344.04 | 1.2785 |
| | $P$ = 0.18 MPa ($T_{sat}$ = −12.73°C) | | | | $P$ = 0.20 MPa ($T_{sat}$ = −10.09°C) | | | | $P$ = 0.24 MPa ($T_{sat}$ = −5.37°C) | | | |
| Sat. | 0.10983 | 219.94 | 239.71 | 0.9273 | 0.09933 | 221.43 | 241.30 | 0.9253 | 0.08343 | 224.07 | 244.09 | 0.9222 |
| −10 | 0.11135 | 222.02 | 242.06 | 0.9362 | 0.09938 | 221.50 | 241.38 | 0.9256 | | | | |
| 0 | 0.11678 | 229.67 | 250.69 | 0.9684 | 0.10438 | 229.23 | 250.10 | 0.9582 | 0.08574 | 228.31 | 248.89 | 0.9399 |
| 10 | 0.12207 | 237.44 | 259.41 | 0.9998 | 0.10922 | 237.05 | 258.89 | 0.9898 | 0.08993 | 236.26 | 257.84 | 0.9721 |
| 20 | 0.12723 | 245.33 | 268.23 | 1.0304 | 0.11394 | 244.99 | 267.78 | 1.0206 | 0.09339 | 244.30 | 266.85 | 1.0034 |
| 30 | 0.13230 | 253.36 | 277.17 | 1.0604 | 0.11856 | 253.06 | 276.77 | 1.0508 | 0.09794 | 252.45 | 275.95 | 1.0339 |
| 40 | 0.13730 | 261.53 | 286.24 | 1.0898 | 0.12311 | 261.26 | 285.88 | 1.0804 | 0.10181 | 260.72 | 285.16 | 1.0637 |
| 50 | 0.14222 | 269.85 | 295.45 | 1.1187 | 0.12758 | 269.61 | 295.12 | 1.1094 | 0.10562 | 269.12 | 294.47 | 1.0930 |
| 60 | 0.14710 | 278.31 | 304.79 | 1.1472 | 0.13201 | 278.10 | 304.50 | 1.1380 | 0.10937 | 277.67 | 303.91 | 1.1218 |
| 70 | 0.15193 | 286.93 | 314.28 | 1.1753 | 0.13639 | 286.74 | 314.02 | 1.1661 | 0.11307 | 286.35 | 313.49 | 1.1501 |
| 80 | 0.15672 | 295.71 | 323.92 | 1.2030 | 0.14073 | 295.53 | 323.68 | 1.1939 | 0.11674 | 295.18 | 323.19 | 1.1780 |
| 90 | 0.16148 | 304.63 | 333.70 | 1.2303 | 0.14504 | 304.47 | 333.48 | 1.2212 | 0.12037 | 304.15 | 333.04 | 1.2055 |
| 100 | 0.16622 | 313.72 | 343.63 | 1.2573 | 0.14932 | 313.57 | 343.43 | 1.2483 | 0.12398 | 313.27 | 343.03 | 1.2326 |
| | $P$ = 0.28 MPa ($T_{sat}$ = −1.23°C) | | | | $P$ = 0.32 MPa ($T_{sat}$ = 2.48°C) | | | | $P$ = 0.40 MPa ($T_{sat}$ = 8.93°C) | | | |
| Sat. | 0.07193 | 226.38 | 246.52 | 0.9197 | 0.06322 | 228.43 | 248.66 | 0.9177 | 0.05089 | 231.97 | 252.32 | 0.9145 |
| 0 | 0.07240 | 227.37 | 247.64 | 0.9238 | | | | | | | | |
| 10 | 0.07613 | 235.44 | 256.76 | 0.9566 | 0.06576 | 234.61 | 255.65 | 0.9427 | 0.05119 | 232.87 | 253.35 | 0.9182 |
| 20 | 0.07972 | 243.59 | 265.91 | 0.9883 | 0.06901 | 242.87 | 264.95 | 0.9749 | 0.05397 | 241.37 | 262.96 | 0.9515 |
| 30 | 0.08320 | 251.83 | 275.12 | 1.0192 | 0.07214 | 251.19 | 274.28 | 1.0062 | 0.05662 | 249.89 | 272.54 | 0.9837 |
| 40 | 0.08660 | 260.17 | 284.42 | 1.0494 | 0.07518 | 259.61 | 283.67 | 1.0367 | 0.05917 | 258.47 | 282.14 | 1.0148 |
| 50 | 008992 | 268.64 | 293.81 | 1.0789 | 0.07815 | 268.14 | 293.15 | 1.0665 | 0.06164 | 267.13 | 291.79 | 1.0452 |
| 60 | 0.09319 | 277.23 | 303.32 | 1.1079 | 0.08106 | 276.79 | 302.72 | 1.0957 | 0.06405 | 275.89 | 301.51 | 1.0748 |
| 70 | 0.09641 | 285.96 | 312.95 | 1.1364 | 0.08392 | 285.56 | 312.41 | 1.1243 | 0.06641 | 284.75 | 311.32 | 1.1038 |
| 80 | 0.09960 | 294.82 | 322.71 | 1.1644 | 0.08674 | 294.46 | 322.22 | 1.1525 | 0.06873 | 293.73 | 321.23 | 1.1322 |
| 90 | 0.10275 | 303.83 | 332.60 | 1.1920 | 0.08953 | 303.50 | 332.15 | 1.1802 | 0.07102 | 302.84 | 331.25 | 1.1602 |
| 100 | 0.10587 | 312.98 | 342.62 | 1.2193 | 0.09229 | 312.68 | 342.21 | 1.1076 | 0.07327 | 312.07 | 341.38 | 1.1878 |
| 110 | 0.10897 | 322.27 | 352.78 | 1.2461 | 0.09503 | 322.00 | 352.40 | 1.2345 | 0.07550 | 321.44 | 351.64 | 1.2149 |
| 120 | 0.11205 | 331.71 | 363.08 | 1.2727 | 0.09774 | 331.45 | 362.73 | 1.2611 | 0.07771 | 330.94 | 362.03 | 1.2417 |
| 130 | | | | | | | | | 0.07991 | 340.58 | 372.54 | 1.2681 |
| 140 | | | | | | | | | 0.08208 | 350.35 | 383.18 | 1.2941 |

（請接第三頁）

類　　科：冷凍空調工程技師
科　　目：冷凍空調設計

附表二：冷媒 R-134a 過熱蒸汽性質表（續）

Superheated refrigerant-134a (*Concluded*)

| *T* °C | *v* m³/kg | *u* kJ/kg | *h* kJ/kg | *s* kJ/kg · K | *v* m³/kg | *u* kJ/kg | *h* kJ/kg | *s* kJ/kg · K | *v* m³/kg | *u* kJ/kg | *h* kJ/kg | *s* kJ/kg · K |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | *P* = 0.50 MPa ($T_{sat}$ = 15.74°C) | | | | *P* = 0.60 MPa ($T_{sat}$ = 21.58°C) | | | | *P* = 0.70 MPa ($T_{sat}$ = 26.72°C) | | | |
| Sat. | 0.04086 | 253.64 | 256.07 | 0.9117 | 0.03408 | 238.74 | 259.19 | 0.9097 | 0.02918 | 241.42 | 261.85 | 0.9080 |
| 20 | 0.04188 | 239.40 | 260.34 | 0.9264 | | | | | | | | |
| 30 | 0.04416 | 248.20 | 270.28 | 0.9597 | 0.03581 | 246.41 | 267.89 | 0.9388 | 0.02979 | 244.51 | 265.37 | 0.9197 |
| 40 | 0.04633 | 256.99 | 280.16 | 0.9918 | 0.03774 | 255.45 | 278.09 | 0.9719 | 0.03157 | 253.83 | 275.93 | 0.9539 |
| 50 | 0.04842 | 265.83 | 290.04 | 1.0229 | 0.03958 | 264.48 | 288.23 | 1.0037 | 0.03324 | 263.08 | 286.35 | 0.9867 |
| 60 | 0.05043 | 274.73 | 299.95 | 1.0531 | 0.04134 | 273.54 | 298.35 | 1.0346 | 0.03482 | 272.31 | 296.69 | 1.0182 |
| 70 | 0.05240 | 283.72 | 309.92 | 1.0825 | 0.04304 | 282.66 | 308.48 | 1.0645 | 0.03634 | 281.57 | 307.01 | 1.0487 |
| 80 | 0.05432 | 292.80 | 319.96 | 1.1114 | 0.04469 | 291.86 | 318.67 | 1.0938 | 0.03781 | 290.88 | 317.35 | 1.0784 |
| 90 | 0.05620 | 302.00 | 330.10 | 1.1397 | 0.04631 | 301.14 | 328.93 | 1.1225 | 0.03924 | 300.27 | 327.74 | 1.1074 |
| 100 | 0.05805 | 311.31 | 340.33 | 1.1675 | 0.04790 | 310.53 | 339.27 | 1.1505 | 0.04064 | 309.74 | 338.19 | 1.1358 |
| 110 | 0.05988 | 320.74 | 350.68 | 1.1949 | 0.04946 | 320.03 | 349.70 | 1.1781 | 0.04201 | 319.31 | 348.71 | 1.1637 |
| 120 | 0.06168 | 330.30 | 361.14 | 1.2218 | 0.05099 | 329.64 | 360.24 | 1.2053 | 0.04335 | 328.98 | 359.33 | 1.1910 |
| 130 | 0.06347 | 339.98 | 371.72 | 1.2484 | 0.05251 | 339.38 | 370.88 | 1.2320 | 0.04468 | 338.76 | 370.04 | 1.2179 |
| 140 | 0.06524 | 349.79 | 382.42 | 1.2746 | 0.05402 | 349.23 | 381.64 | 1.2584 | 0.04599 | 348.66 | 380.86 | 1.2444 |
| 150 | | | | | 0.05550 | 359.21 | 392.52 | 1.2844 | 0.04729 | 358.68 | 391.79 | 1.2706 |
| 160 | | | | | 0.05698 | 369.32 | 403.51 | 1.3100 | 0.04857 | 368.82 | 402.82 | 1.2963 |
| | *P* = 0.80 MPa ($T_{sat}$ = 31.33°C) | | | | *P* = 0.90 MPa ($T_{sat}$ = 35.53°C) | | | | *P* = 1.00 MPa ($T_{sat}$ = 39.39°C) | | | |
| Sat. | 0.02547 | 243.78 | 264.15 | 0.9066 | 0.02255 | 245.88 | 266.18 | 0.9054 | 0.02020 | 247.77 | 267.97 | 0.9043 |
| 40 | 0.02691 | 252.13 | 273.66 | 0.9374 | 0.02325 | 250.32 | 271.25 | 0.9217 | 0.02029 | 248.39 | 268.68 | 0.9066 |
| 50 | 0.02846 | 261.62 | 284.39 | 0.9711 | 0.02472 | 260.09 | 282.34 | 0.9566 | 0.02171 | 258.48 | 280.19 | 0.9428 |
| 60 | 0.02992 | 271.04 | 294.98 | 1.0034 | 0.02609 | 269.72 | 293.21 | 0.9897 | 0.02301 | 268.35 | 291.36 | 0.9768 |
| 70 | 0.03131 | 280.45 | 305.50 | 1.0345 | 0.02738 | 279.30 | 303.94 | 1.0214 | 0.02423 | 278.11 | 302.34 | 1.0093 |
| 80 | 0.03264 | 289.89 | 316.00 | 1.0647 | 0.02861 | 288.87 | 314.62 | 1.0521 | 0.02538 | 287.82 | 313.20 | 1.0405 |
| 90 | 0.03393 | 299.37 | 326.52 | 1.0940 | 0.02980 | 298.46 | 325.28 | 1.0819 | 0.02649 | 297.53 | 324.01 | 1.0707 |
| 100 | 0.03519 | 308.93 | 337.08 | 1.1227 | 0.03095 | 308.11 | 335.96 | 1.1109 | 0.02755 | 307.27 | 334.82 | 1.1000 |
| 110 | 0.03642 | 318.57 | 347.71 | 1.1508 | 0.03207 | 317.82 | 346.68 | 1.1392 | 0.02858 | 317.06 | 345.65 | 1.1286 |
| 120 | 0.03762 | 328.31 | 358.40 | 1.1784 | 0.03316 | 327.62 | 357.47 | 1.1670 | 0.02959 | 326.93 | 356.52 | 1.1567 |
| 130 | 0.03881 | 338.14 | 369.19 | 1.2055 | 0.03423 | 337.52 | 368.33 | 1.1943 | 0.03058 | 336.88 | 367.46 | 1.1841 |
| 140 | 0.03997 | 348.09 | 380.07 | 1.2321 | 0.03529 | 347.51 | 379.27 | 1.2211 | 0.03154 | 346.92 | 378.46 | 1.2111 |
| 150 | 0.04113 | 358.15 | 391.05 | 1.2584 | 0.03633 | 357.61 | 390.31 | 1.2475 | 0.03250 | 357.06 | 389.56 | 1.2376 |
| 160 | 0.04227 | 368.32 | 402.14 | 1.2843 | 0.03736 | 367.82 | 401.44 | 1.2735 | 0.03344 | 367.31 | 400.74 | 1.2638 |
| 170 | 0.04340 | 378.61 | 413.33 | 1.3098 | 0.03838 | 378.14 | 412.68 | 1.2992 | 0.03436 | 377.66 | 412.02 | 1.2895 |
| 180 | 0.04452 | 389.02 | 424.63 | 1.3351 | 0.03939 | 388.57 | 424.02 | 1.3245 | 0.03528 | 388.12 | 423.40 | 1.3149 |

附圖一：空氣線圖

